

Weichi Yao

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Education

- 2026-** **Postdoctoral Research Fellow**, Statistics Department - University of Michigan
AI Mentor Prof. Yixin Wang
Science Mentor Prof. Bryan R. Goldsmith
- 2023-2025** **Postdoctoral Research Fellow**, Schmidt AI in Science - University of Michigan
AI Mentor Prof. Yixin Wang
Science Mentor Prof. Bryan R. Goldsmith
- 2017-2023** **Ph.D.**, Statistics - New York University
M.Phil, Statistics
Advisor Prof. Halina Frydman
Thesis: Machine Learning for Science
- 2015-2016** **M.A.**, Statistics - Columbia University
- 2011-2015** **B.A.**, Finance - Wuhan University
B.A., Applied Mathematics

Grants

- 2025-2027** ACED: Tail-aware Generative Modeling for Inverse Discovery of Molecules (Co-PI).
National Science Foundation (Total: \$500K).
PI: Y. Wang (Statistics); Co-PI: B. R. Goldsmith (Chemical Engineering).
- 2025-2026** Reasoning-Augmented Large Language Models for Molecule Design and Discovery.
University of Michigan-OpenAI Research Grants (Total: \$50K).
PI: B. R. Goldsmith (Chemical Engineering).

Ongoing Projects

1. W. Yao, B. R. Goldsmith and Y. Wang.
Closed-Loop Multi-Fidelity Molecular Design and Property Prediction.
Key contribution: Developing a closed-loop framework that integrates molecular generation, property prediction, and computational validation across multiple levels of fidelity. Predictions guide candidate generation, while high-fidelity evaluation of informative candidates improves targeted molecular discovery. This reciprocal feedback progressively improves both property prediction accuracy and reliability of discovering molecules that satisfy specified targets.

2. W. Yao, C. Gruich, B. R. Goldsmith and Y. Wang.
Equivariant-Free Transformer-Autoencoded Latent Flow Matching for 3D Molecule Generation.
Key contribution: (1) a fixed-d latent design that lets the autoencoder and downstream flow matching model jointly determine composition, geometry, and size, enabling open-ended autoregressive generation without committing to a molecule size in advance; (2) a lightweight preprocessing step that captures the practical benefits of SE(3) invariance handling without SE(3) equivariant layers or data augmentation.
3. W. Yao, B. R. Goldsmith and Y. Wang.
Extrapolative conditional molecule generation.
Key contribution: Building a lightweight transformer adapter for pretrained conditional generative model, enabling reliable generation when target properties outside the training distribution without retraining the base model. Validated the method on synthetic benchmarks and QM9 molecular inverse-design tasks, improving out-of-distribution accuracy and extrapolative robustness over the baseline generator.
4. D. Sweeney, W. Yao, Y. Wang and B. R. Goldsmith
De Novo generation of catalyst/adsorbate configurations with target binding energies.
Key contribution: Developing a flow matching model with transformer as backbone, which generates complete catalyst-adsorbate configurations from scratch, including catalyst structures, adsorbate atom types, and optimized adsorption geometries, conditioned on target binding energies.

Preprints

- 2026 W. Yao, B. Dumitrascu, B. R. Goldsmith and Y. Wang. Goal-Oriented Influence-Maximizing Data Acquisition for Learning and Optimization. *Submitted to Journal of Machine Learning Research.*
- 2024 W. Yao, C. Gruich, B. R. Goldsmith and Y. Wang. Tail extrapolative conditional molecule generation. *In Proceedings of the International Conference on Machine Learning (ICML) AI4Science Workshop.*
- 2021 W. Yao, K. Storey-Fisher, D. W. Hogg and S. Villar. A simple equivariant machine learning method for dynamics based on scalars. *In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS) Workshop on Machine Learning and the Physical Sciences.*

Publications

- 2024 S. Villar, D. W. Hogg, W. Yao, G. A. Kevrekidis and B. Schölkopf. Towards fully covariant machine learning. *Transactions on Machine Learning Research.*
- 2023 S. Villar, W. Yao, D. W. Hogg, B. Blum-Smith and B. Dumitrascu. Dimensionless machine learning: Imposing exact units equivariance. *Journal of Machine Learning Research*, 24.
- 2022 W. Yao, H. Frydman, D. Larocque and J. S. Simonoff. Ensemble methods for survival function estimation with time-varying covariates. *Statistical Methods in Medical Research*, 31(11): 2217-2236.
- 2021 W. Yao, H. Frydman and J. S. Simonoff. An ensemble method for interval-censored time-to-event data. *Biostatistics*, 22(1): 198-213.

- 2021** H. Moradian, W. Yao, D. Larocque, J. S. Simonoff and H. Frydman. Dynamic estimation with random forests for discrete-time survival data. *The Canadian Journal of Statistics*, 50(2): 533-548.
- 2021** S. Villar, D. W. Hogg, K. Storey-Fisher, W. Yao and B. Blum-Smith. Scalars are universal: Equivariant machine learning, structured like classical physics. *In Proceedings of the Advances in Neural Information Processing Systems*
- 2019** W. Yao, A. S. Bandeira and S. Villar. Experimental performance of graph neural networks on random instances of max-cut. *In Proceedings of the Society of Photographic Instrumentation Engineers*.
- 2017** J. H. Lee, D. E. Carlson, H. S. Razaghi, W. Yao, G. A. Goetz, E. Hagen, E. Batty, E. J. Chichilnisky, G. T. Einevoll and L. Paninski. YASS: Yet Another Spike Sorter. *In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS)*, 4005-4015.

Internship

- 12/2022-05/2023** Science Intern. Amazon Development Center, Germany
Transfer learning with deep tabular models.
- 05/2022-09/2022** Research Intern. Applied Machine Learning Team, TikTok at ByteDance
Maximum likelihood estimation with full-likelihood formulation for click-through rate prediction on nonuniform subsampled data.
- 12/2016-07/2017** Research Assistant. Grossman Center for the Statistics of Mind, Columbia University
Statistical models and pipelines for spike sorting on dense multi-electrode arrays.

Teaching experience

- 2020** Course Instructor. STAT-UB Statistics for Business Control, New York University.

Prizes or Awards

- 2014** Honorable Mention. Mathematical Contest in Modeling (MCM)
- 2013** Second Prize. Symphony Orchestra, National College Students Art Exhibition, China.
- 2009** Gold Award. Symphony Orchestra, Australian International Music Festival, Australia.